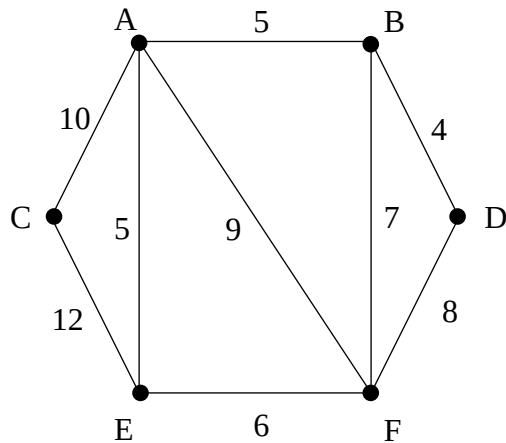


Discrete Mathematics: Network Algorithms revision

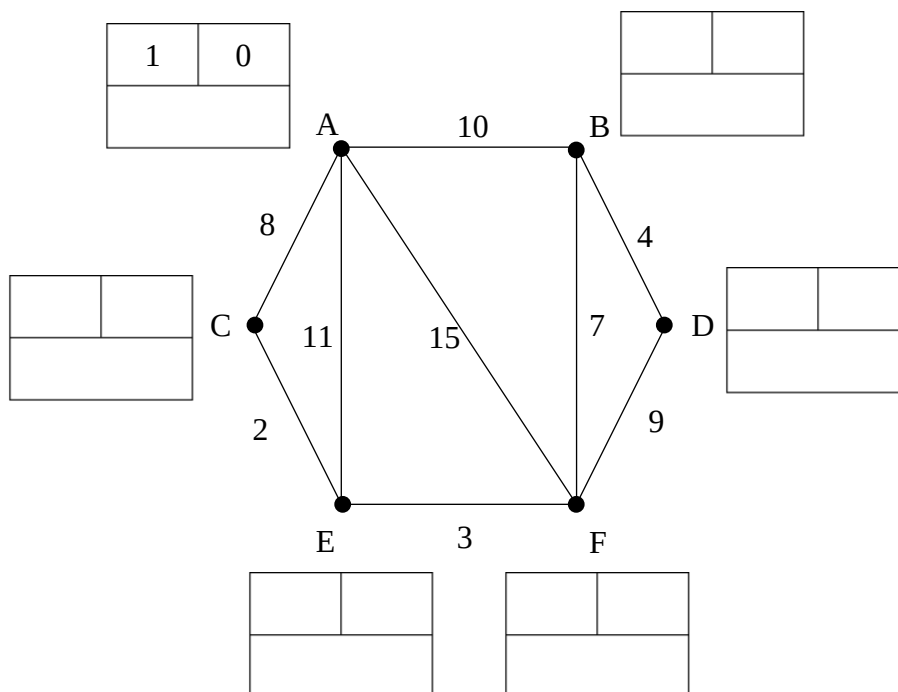
1 Show how to find a minimum spanning tree of the network shown below using:

(a) Prim's algorithm, starting at A.

(b) Kruskal's algorithm



2 Use Dijkstra's algorithm to find the shortest path from A to F in the network shown below



- 3 Show how to use the tabular version of Prim's algorithm to find the minimum spanning tree of the network described the distance matrix below.

	A	B	C	D	E	F
A	-	8	10	-	12	5
B	8	-	6	18	2	-
C	10	6	-	7	-	9
D	-	18	7	-	6	-
E	12	2	-	6	-	3
F	5	-	9	-	3	-

- 4 Duke has written a computer program to implement the tabular version of Prim's algorithm. When he inputs a 50×50 distance matrix it takes his programme 78 milliseconds to output the minimum spanning tree. Estimate how long his program would take if he inputted a 200×200 distance matrix.
- 5 Given a network on n vertices, consider repeatedly using Dijkstra's algorithm to find the shortest distance between every pair of vertices. If it takes 2.4 seconds for a computer to do this for a complete network on 35 vertices, then roughly how long will it take the same computer do to this for a complete network on 100 vertices?
- 6 The weights on the edges of a complete network are sorted into ascending order using bubble sort. It takes a computer 182 milliseconds to do this for a complete network on 60 vertices. Roughly how long will it take the same computer to do this for a complete network on 500 vertices?